INCH-POUND
MIL-PRF-49462C
w/AMENDMENT 5
6 January 2026
SUPERSEDING
MIL-PRF-49462C
w/AMENDMENT 4
28 October 2019

PERFORMANCE SPECIFICATION

RESISTOR, FIXED, FILM, HIGH VOLTAGE
GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments
and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for fixed, film, high voltage resistors primarily intended for incorporation into electronic circuits where high voltages and high resistance values are present.

1.2 Classification.

1.2.1 Part or Identifying Number (PIN). The PIN is in the following form, and as specified (see 3.1).

M4946203	A	A	100K	F
Specification and associated slash sheet	Characteristic (1.2.1.1)	Style (1.2.1.2)	Resistance (1.2.1.3)	Tolerance (1.2.1.4)

1.2.1.1 Characteristic. The characteristic is identified by a single letter A, in accordance with table I.

1.2.1.2 Style. The style is identified by a single letter (see 3.1). The styles available are A, B, C, D, E, and F.

1.2.1.3 Resistance. The nominal resistance expressed in ohms is identified by four characters, consisting of three digits and a letter. The letter is used simultaneously as a decimal point and as a multiplier. For resistance values:

a. Greater than or equal to 1 ohm but less than 1 kilohm, the letter “R” is used to represent a decimal point.
b. Greater than or equal to 1 kilohm but less than 1 megohm, the letter “K” is used to represent a decimal point.
c. Greater than or equal to 1 megohm but less than 1 gigohm, the letter “M” is used to represent a decimal point.
d. Equal to 1 gigohm, the letter “G” is used to represent a decimal point.

All digits proceeding and following the letter (R, K, M, G) of the group represent significant figures. The resistance values designations are shown in table II. The minimum and maximum are as specified herein and 3.1. The standard values for every decade follows the sequence specified in table III.

Comments, suggestions, or questions on this document should be addressed to: DLA Weapons Support (Columbus), ATTN: VAT, Post Office Box 3990, Columbus, Ohio 43218-3990 or by email Resistor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

AMSC N/A
FSC 5905
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:08Z
Check the source to verify that this is the current version before use.

TABLE I. Characteristics.

Test or condition	Characteristic A	Units
Maximum resistance temperature characteristic (RTC)	Resistance / RTC < 500 MΩ / 200 ≥ 500 MΩ / 500	ppm/°C
Maximum ambient temperature at rated power.	+70	°C
Maximum ambient temperature at zero wattage derating.	+175	
Power rating.		Watts
RHV30	0.25	
RHV31	0.5	
RHV32	1.0	
RHV33	2.0	
RHV34	3.0	
RHV35	5.0	
Maximum dc or rms voltage		Volts
RHV30	750	
RHV31	1,500	
RHV32	3,000	
RHV33	5,000	
RHV34	10,000	
RHV35	20,000	
Thermal shock	±0.5	Maximum percent change in resistance
Dielectric withstanding voltage	±0.25	
Life	±5.0	
Shock (specified pulse)	±2.0	
Terminal strength	±0.2	
Vibration, high frequency	±2.0	
Low temperature storage	±0.25	
Resistance tolerance	F (±1), G (±2), J (±5)	Percent

TABLE II. Designation of resistance.

Designation			Resistance (ohms)		
10R0	to	97R6	10	to	97.6
100R	to	976R	100	to	976
1K00	to	9K76	1,000	to	9,760
10K0	to	97K6	10,000	to	97,600
100K	to	976K	100,000	to	976,000
1M00	to	9M76	1,000,000	to	9,760,000
10M0	to	97M6	10,000,000	to	97,600,000
100M	to	976M	100,000,000	to	976,000,000
1G00			1,000,000,000		

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:08Z
Check the source to verify that this is the current version before use.

TABLE III. Standard resistance values for the 10 decade to 100 decade.

Tolerance									
F (±1.0)	G (±2.0) J (±5.0)	F (±1.0)	G (±2.0) J (±5.0)	F (±1.0)	G (±2.0) J (±5.0)	F (±1.0)	G (±2.0) J (±5.0)
10.00	10.00	18.20		33.20		56.20	
10.20		18.70		34.00		57.60	
10.50		19.10		34.80		59.00	
10.70		20.00	20.00	35.70		60.40	
11.00	11.00	20.50			36.00	61.90	
11.30		21.00		36.50			62.00
11.50		21.50		37.40		63.40	
11.80			22.00	38.30		64.90	
	12.00	22.10			39.00	66.50	
12.10		22.60		39.20			68.00
12.40		23.20		40.20		68.10	
12.70		23.70		41.20		69.80	
13.00	13.00		24.00	42.20		71.50	
13.30		24.30			43.00	73.20	
13.70		24.90		43.20		75.00	75.00
14.00		25.50		44.20		76.80	
14.30		26.10		45.30		78.70	
14.70		26.70		46.40		80.60	
15.00	15.00		27.00		47.00		82.00
15.40		27.40		47.50		82.50	
15.80		28.00		48.70		84.50	
	16.00	28.70		49.90		86.60	
16.20		29.40			51.00	88.70	
16.50			30.00	51.10		90.90	
16.90		30.10		52.30			91.00
17.40		30.90		53.60		93.10	
17.80		31.60		54.90		95.30	
	18.00	32.40			56.00	97.60	
			33.00				

1.2.1.4 Resistance tolerance. The resistance tolerance is identified by a single letter in accordance with table IV.

TABLE IV. Resistance tolerance. 1/

Symbol	Resistance tolerance in percent
F	±1.0
G	±2.0
J	±5.0

1/ Minimum and maximum resistance values may vary with tolerances (see 3.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:08Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

DEPARTMENT OF DEFENSE SPECIFICATION

MIL-PRF-49462/3	-	Resistor, Fixed, Film, High Voltage, Styles RHV30, RHV31, RHV32, RHV33, RHV34, RHV35.

DEPARTMENT OF DEFENSE STANDARD

MIL-STD-202	-	Electronics and Electrical Components Parts.
MIL-STD-202-105	-	Method 105, Barometric Pressure (Reduced)
MIL-STD-202-106	-	Method 106, Moisture Resistance
MIL-STD-202-107	-	Method 107, Thermal Shock
MIL-STD-202-108	-	Method 108, Life (At Elevated Ambient Temperature)
MIL-STD-202-204	-	Method 204, Vibration, High Frequency
MIL-STD-202-208	-	Method 208, Solderability
MIL-STD-202-211	-	Method 211, Terminal Strength
MIL-STD-202-213	-	Method 213, Shock (Specified Pulse)
MIL-STD-202-301	-	Method 301, Dielectric Withstanding Voltage
MIL-STD-202-302	-	Method 302, Insulation Resistance
MIL-STD-202-303	-	Method 303, DC Resistance
MIL-STD-202-304	-	Method 304, Resistance-Temperature Characteristic
MIL-STD-1285	-	Marking of Electrical and Electronic Parts.

(Copies of these documents are available online at https://quicksearch.dla.mil).

2.3 Non-government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are those listed in the solicitation or contract.

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 10012	-	Measurement Management Systems - Requirements for Measurement Processes and Measuring Equipment
ISO/IEC 17025	-	Competence of Testing and Calibration Laboratories, General requirements for the

(Copies of this document are available online at https://www.iso.org)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:08Z
Check the source to verify that this is the current version before use.

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence unless otherwise noted. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 Qualification. Qualification is not required.

3.3 Material. The material shall be as specified herein. However, when a definite material is not specified, a material shall be used which will enable the resistors to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product. Material shall be corrosion resistant or shall be processed to resist corrosion.

3.4 Interface and physical dimension requirements. Resistors shall meet the interface and physical dimensions as specified (see 3.1).

3.4.1 Terminals. Terminal leads shall be free of foreign material and solderable in the clean lead to clean lead area. Leads shall be judged to be free of foreign material if the visual criteria are met (see 3.9).

3.4.2 Films. The films shall be uniformly deposited. The film shall be free of blisters, thin spots, areas inadequately bonded to the core, discolored spots or other blemishes likely to cause flaking or a nonuniform ribbon when spiraled (helixed). Where used, spiraling shall occupy no less than 70 percent of the resistor element actual length. The resistor element actual length shall be defined as the nominal distance between terminal bands less 0.046875 inch.

3.4.2.1 Carbon films. Carbon films shall not be used.

3.4.3 End caps. When end caps are used in construction of the resistor, the misalignment of the cap with respect to the core shall not exceed 5 degrees.

3.5 Power rating. Power rating is based on continuous full load operation at a rated ambient temperature. See the applicable specification sheet (see 3.1) for the applicable derating curve.

3.6 Voltage rating. Resistor shall have a rated dc continuous working voltage or an approximate sine-wave root mean square (rms) continuous working voltage corresponding to the wattage (power) rating, as determined from the following formula:

$$E = \sqrt{PR}$$

Where: E = Continuous rated dc or rms working voltage in volts.
P = Rated wattage in watts.
R = Nominal resistance in ohms.

In no case shall the rated voltage be greater than the applicable maximum voltage.

3.7 DC resistance. When resistors are tested as specified in 4.6.2, the dc resistance shall be within the specified tolerance of the nominal resistance (see 1.2.1.3), and shall be stable within the specified tolerance during the measurement.

3.7.1 Resistance value deviations. All maximum deviations as specified in this section are to be considered absolute limits with the exception of the contact resistance adjustments.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:08Z
Check the source to verify that this is the current version before use.

3.8 Power conditioning. When resistors are tested as specified in 4.6.3, there shall be no evidence of mechanical damage and the change in resistance shall not exceed ±0.5 percent.

3.9 Solderability. The dipped surface of the leads of the resistor tested as specified in 4.6.4, shall be at least 95 percent covered with a new, smooth coating and shall exhibit no demetallization or leaching of terminal areas. The remaining 5 percent may contain only pin holes or rough spots, which shall not be concentrated in one area. In case of dispute, the percentage of coverage with pinholes or rough spots shall be determined by actual measurement of these areas, as compared to the total area.

3.10 Thermal shock. When resistors are tested as specified in 4.6.5 , there shall be no evidence of mechanical damage. The change in resistance shall not exceed ±0.5 percent.

3.11 Low temperature storage. When resistors are tested as specified in 4.6.6, there shall be no evidence of mechanical damage and the change in resistance between the initial and final measurement at +25°C ±5°C shall not exceed ±(0.25 percent +0.05 ohm).

3.12 Terminal strength. When resistors are tested as specified in 4.6.7, there shall be no evidence of breaking or chipping of the coating. The change in resistance shall not exceed ±0.2 percent.

3.13 Pure tin. The use of pure tin, as an underplate or final finish is prohibited both internally and externally. Tin content of resistor components and solder shall not exceed 97 percent, by mass. Tin shall be alloyed with a minimum of 3 percent lead, by mass (see 6.4).

3.14 Solder dip (retinning) leads. The manufacturer may solder dip/retin the leads of product supplied to this specification provided the solder dip/retin process has been approved by the acquiring activity.

3.15 Dielectric withstanding voltage. When resistors are tested as specified in 4.6.8, there shall be no evidence of flashover, mechanical damage, arcing, or insulation breakdown. The change in resistance shall not exceed ±0.25 percent.

3.16 Insulation resistance. When resistors are tested as specified in 4.6.9, the insulation resistance shall not be less than 10,000 megohms.

3.17 Moisture resistance. When resistors are tested as specified in 4.6.10, there shall be no evidence of mechanical damage. The change in resistance between initial and final measurements shall not exceed ±1.00 percent. In addition, the dielectric withstanding voltage shall be as specified in 3.15, and insulation resistance shall be 100 megohms, minimum.

3.18 Life. When resistors are tested as specified in 4.6.11, there shall be no evidence of mechanical damage. The change in resistance between initial measurement and any succeeding measurements shall not exceed 5 percent.

3.19 Resistance temperature characteristic. When resistors are tested as specified in 4.6.12, the resistance temperature characteristic, at each of the temperature specified in 4.6.12b referred to room ambient temperature, shall not exceed 200 ppm/^{O}C for resistance values less than 500 megohms and 500 ppm/^{O}C for resistance values greater than or equal to 500 megohms.

3.20 Shock, specified pulse. When resistors are tested as specified in 4.6.13, there shall be no evidence of mechanical damage or electrical damage. The change in resistance shall not exceed ±2.0 percent and there shall be no electrical discontinuity during testing.

3.21 Vibration, high frequency. When resistors are tested as specified in 4.6.14 , there shall be no evidence of mechanical damage. The change in resistance shall not exceed ±2 percent and there shall be no electrical discontinuity during the test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:08Z
Check the source to verify that this is the current version before use.

3.22 Marking. Each resistor shall be marked in accordance with method I of MIL-STD-1285 and as indicated below:

a. Manufacturer's code symbol.
b. PIN.
c. Manufacturing data code.

3.23 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.24 Workmanship. Resistors shall be processed in such a manner as to be uniform in quality and shall be free from any defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. Conformance inspection (see 4.5).

4.2 Test equipment and inspection facilities. The manufacturer shall establish and maintain a calibration system in accordance with ISO/IEC 17025, ISO 10012, or equivalent system as approved by the acquiring activity.

4.3 Inspection conditions and precautions.

4.3.1 Inspection conditions. Unless otherwise specified herein, all inspections shall be in accordance with the test conditions specified in the GENERAL REQUIREMENTS of MIL-STD-202.

4.3.2 Precautions. Adequate precautions shall be taken during tests to prevent condensation of moisture on resistors, except during the moisture resistance tests, and thermal shock tests. Precaution shall also be taken to prevent damage by heat when soldering resistor leads to terminals.

4.4 Qualification inspection. Qualification inspection is not required.

4.5 Conformance inspection.

4.5.1 Inspection of product for delivery. Inspection of product for delivery shall consist of group A, group B, and group C inspections.

4.5.1.1 Inspection and production lot.

4.5.1.1.1 Inspection lot. An inspection lot, as far as practicable, shall consist of all resistors of the same style, characteristic, and protective enclosure or coating produced in a period not to exceed 30 days, produced under essentially the same conditions and offered for inspection at one time.

4.5.1.1.2 Production lot. A production lot shall consist of all resistors of the same characteristic, style, nominal resistance value, and resistance tolerance. Manufacture of all parts in the lot shall have been started, processed, assembled, and tested as a group. Lot identity shall be maintained throughout the manufacturing process.

4.5.1.2 Certification.. The acquiring activity, at its discretion, may accept a certificate of compliance with group C requirements in lieu of performing group C tests (see 6.2d). NOTE: Group C testing includes a 1,000-hour life test which may extend the processing time by 45 days or more.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:08Z
Check the source to verify that this is the current version before use.

4.5.2 Group A inspection. Group A inspection shall consist of the inspections specified in table V and shall be made on the same set of sample units, in the order shown.

4.5.2.1 Sampling plan.

4.5.2.1.1 Subgroup I. Subgroup 1 tests shall be performed on a production lot basis on 100 percent of the product supplied under this specification. Resistors that are out of resistance tolerance, or which experience a change in resistance greater than that permitted for the tests of this subgroup shall be removed from the lot. Lots having more than 10 percent total rejects, due to exceeding the specified resistance change limit shall not be furnished on contracts.

TABLE V. Group A inspection.

Inspection	Requirement paragraph	Method paragraph	Number of samples
Subgroup I			
Thermal shock	3.10	4.6.5	100 percent inspection
Power conditioning	3.8	4.6.3	
DC resistance	3.7	4.6.2	
Subgroup II			
Visual and mechanical examination			
Body dimension	3.1		
Diameter and length of leads	3.1	4.6.1	See 4.5.2.1.2
Marking 1/	3.22		
Workmanship	3.24		
Subgroup III			
Solderability	3.9	4.6.4	See 4.5.2.1.3

1/ Marking defects shall be charged only for illegible, incorrect, or incomplete marking. Any subsequent electrical defect shall not be charged as a marking defect.

4.5.2.1.2 Subgroup II. A sample of parts from each inspection lot shall be randomly selected in accordance with table VI, if one or more defects are found, the lot shall be rescreened and defects removed. After screening and removal of defects, a new sample of parts shall be randomly selected in accordance with table VI, if one or more defects are found in the second sample, the lot shall be rejected and shall not be supplied to this specification.

TABLE VI. Group A sampling plan.

Lot size			Subgroup II sample size
2	to	13	100%
14	to	125	13
126	to	150	13
151	to	280	20
281	to	500	29
501	to	1,200	34
1,201	to	3,200	42
3,201	to	10,000	50
10,001	to	35,000	60
35,001	to	150,000	74
150,001	to	500,000	90
500,001	and over		102

4.5.2.1.3 Subgroup III (solderability).

4.5.2.1.3.1 Sampling plan. Thirteen samples shall be selected randomly from each inspection lot and subjected to the subgroup III solderability test. The manufacturer may use electrical rejects from the subgroup I screening tests for all or part of the samples to be used for solderability testing. If there are one or more defects, the lot shall be considered to have failed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:08Z
Check the source to verify that this is the current version before use.

4.5.2.1.3.2 Rejected lots. In the event of one or more defects, the inspection lot is rejected. The manufacturer may use of the following options to rework the lot:

a. Each production lot that was used to form the failed inspection lot shall be individually submitted to the solderability test as required in 4.6.4. Production lots that pass the solderability tests are available for shipment. Production lots failing the solderability test can be reworked and resubmitted to the sample plan. If one or more defects are found in this second sample, the lot is rejected and shall not be supplied to this specification.

4.5.2.1.3.3 Disposition of samples. The solderability test is considered a destructive test and samples submitted to the solderability test shall not be supplied on the contract.

4.5.3 Group B inspection. Group B inspection shall consist of the inspections specified in table VII in the order shown. They shall be performed on sample units that have been subjected to and passed the group A inspection.

TABLE VII. Group B inspection

Inspection	Requirement paragraph	Method paragraph
Resistance temperature characteristic	3.19	4.6.12
Dielectric withstanding voltage	3.15	4.6.8
Thermal shock	3.10	4.6.5

4.5.3.1 Sampling plan. Group B tests shall be performed on an inspection lot basis. A sample of 13 parts shall be randomly selected. If one or more defects are found, the lot shall be rescreened and defects removed. A new sample of 13 parts shall be randomly selected. If one or more defects are found in this second sample, the lot shall be rejected and shall not be supplied to this specification.

4.5.4 Group C inspection. Group C inspection shall consist of the tests specified in table VIII in the order shown. Group C inspection shall be performed on sample units of each style and selected from inspection lots which have passed group A and group B inspections. The number of sample units to be inspected shall be as specified in table VIII. In the event of failures in excess of those allowed in table VIII, the lot shall be rejected. Parts may not be shipped until representative samples of the lot have successfully completed the group C tests.

TABLE VIII. Group C inspection.

Inspection	Requirement Paragraph	Method paragraph	Number of samples	Number of failures allowed
Thermal shock	3.10	4.6.5	10	1
Low temperature storage	3.11	4.6.6		
Terminal strength	3.12	4.6.7		
Subgroup I				
Dielectric withstanding voltage	3.15	4.6.8		1
Insulation resistance	3.16	4.6.9	10	
Moisture resistance	3.17	4.6.10		
Subgroup II				
Life	3.18	4.6.10	10	1
Shock (specific pulse)	3.20	4.6.13	10	1
Vibration, high frequency	3.21	4.6.14		

4.5.4.1 Disposition of samples. Sample units which have been subjected to group C inspection shall not be delivered on the contract or purchase order.

4.5.4.2 Rejected lots. Rejected lots shall not be supplied to this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:08Z
Check the source to verify that this is the current version before use.

4.5.5 Inspection of packaging. The sampling and inspection of the preservation, packing, and container marking shall be in accordance with the requirements of section 5.

4.6 Methods of inspection.

4.6.1 Visual and mechanical examination. Resistors shall be examined to verify that the materials, design, construction, physical dimensions, marking, and workmanship are in accordance with the applicable requirements (see 3.1 and 3.4 to 3.4.3, 3.22, and 3.24 inclusive).

4.6.2 DC resistance. The dc resistance shall be measured in accordance with MIL-STD-202-303 (see 3.7). The following details and exceptions shall apply:

a. Measuring apparatus: Different types of measuring test equipment (multimeters, bridges, or equivalent) are permitted to be used on the initial and final readings of this test, provided the equipment is the same style, model, or it can be shown that the performance of the equipment is equivalent or better.
b. Test voltage: Measurements of resistance shall be made by using the test voltages specified in table IX. The test voltage chosen, whether it be the maximum or a lower voltage which would still provide the sensitivity required, shall be applied across the terminals of the resistor. This same voltage shall be used whenever a subsequent resistance measurement is made.
c. Temperature: The dc resistance test specified in group I of table V shall be performed at +25°C ±2°C. For all other tests, unless otherwise specified herein, the temperature at which subsequent and final resistance measurements are made in each test shall be within ±2°C of the temperature at which the initial resistance measurement was made.
d. Humidity: Humidity shall be 50 percent.

TABLE IX. DC resistance test voltages.

Resistance, nominal (ohms)	Maximum test voltage (volts)
100 to 976 inclusive	3
1,000 to 9,760 inclusive	10
10,000 to 97,600 inclusive	30
100,000 or higher	100

4.6.3 Power conditioning (see 3.8).

4.6.3.1 Mounting. Resistors may be mounted in any position and allotted as much space as necessary. Forced air cooling may be used to maintain a test ambient temperature of +20°C to +45°C. The velocity of the forced air shall not exceed 500 feet per minute and there shall be no direct impingement of the forced air upon the resistors.

4.6.3.2 Procedure. The load shall be 1.5 times rated power for a duration of 100 hours ±8 hours, not to exceed the maximum voltage specified (see 3.1). The voltage applied may be rectified ac or dc. DC resistance shall be measured before and after the test following a 2 hour stabilization period at +25°C ±2°C. Resistors shall be examined for evidence of arcing, burning, or charring.

4.6.4 Solderability. Resistors shall be tested in accordance with MIL-STD-202-208 (see 3.9) . Both leads shall be tested.

4.6.5 Thermal shock. Resistors shall be tested in accordance with MIL-STD-202-107 (see 3.10). The following details and exceptions shall apply:

a. Mounting: Resistors shall be mounted by means other than soldering.
b. Measuring before cycling: DC resistance shall be measured as specified in 4.6.2.
c. Test condition letter: B, except that the extreme high temperature shall be +150°C +3°C, -0°C.
d. Measurement after cycling: Within 3 hours after completion of the final cycle and as soon as the resistors stabilize at room temperature, dc resistance shall again be measured as specified in 4.6.2.
e. Examination after test: Resistors shall be examined for evidence of mechanical damage.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:08Z
Check the source to verify that this is the current version before use.

4.6.6 Low temperature storage (see 3.11).

4.6.6.1 Mounting. Resistors shall be mounted in such a manner that there is at least 1 inch of free air space around each resistor, and in such a position with respect to the air stream that the mounting offers substantially no obstruction to the flow of air across and around the resistors.

4.6.6.2 Procedure. DC resistance shall be measured as specified in 4.6.2. Within 1 hour after this measurement, the resistors shall be placed in a cold chamber at a temperature of -65°C ±2^{O}C for a period of 24 hours ±4 hours. The resistors shall then be removed from the chamber and maintained at a temperature of +25°C ±5°C for a period of approximately 2 hours to 8 hours; the dc resistance shall again be measured as specified in 4.6.2. Resistors shall then be examined for evidence of mechanical damage.

4.6.7 Terminal strength (see 3.12). Resistors shall be tested in accordance with MIL-STD-202-211. The following details and exceptions shall apply:

a. Test condition: C (wire-lead bend test).
b. Measurement after test: DC resistance shall be measured as specified in 4.6.2.
c. The applied load shall be 1 pounds(16 ounces) ± 1 ounce for all styles
d. Examination after test: DC resistance shall be measured as specified in 4.6.2. Resistors shall be examined for evidence of mechanical damage (see 3.12).

4.6.8 Dielectric withstanding voltage (see 3.15).

4.6.8.1 Atmospheric pressure. Resistors shall be tested in accordance with MIL-STD-202-301. The following details and exceptions shall apply:

a. Mounting: Resistors shall be clamped in the trough of a 90° Metallic V-block of such size that the body of the resistor does not extend beyond the extremities of the block. The resistor leads shall be so positioned that the distance between the resistor lead and any point of the V-block is not less than the radius of the resistor minus the radius of the lead wire.
b. Initial measurement: DC resistance shall be measured as specified in 4.6.2.
c. Nature of potential: Alternating current (ac) supply at commercial line frequency (not more than 100 hertz and waveform).
d. Magnitude of test potential: Sine wave test potential of 900 volts rms.
e. Duration of application of test voltages: 1 minute.
f. Rate of application of test voltage: 100 volts per second.
g. Points of application of test voltage: Between the resistor terminals connected together and the V-block.
h. Measurement after test: DC resistance shall be measured as specified in 4.6.2.
i. Examination after test: Resistors shall be examined for evidence of flashover, mechanical damage, arcing, and insulation breakdown.

4.6.8.2 Barometric pressure (reduced). Resistors shall be tested in accordance with MIL-STD-202-105. The following details and exceptions shall apply:

a. Method of mounting: As specified in 4.6.8.1a.
b. Initial measurement: Dc resistance shall be measured as specified in 4.6.2.
c. Test voltage during subjection to reduced pressure: 450 volts rms.
d. Test condition letter: B.
e. Nature of potential: As specified in 4.6.8.1c.
f. Duration of test: 1 minute.
g. Points of application of test voltage: As specified in 4.6.8.1g.
h. Measurements after test: DC resistance shall be measured as specified in 4.6.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:08Z
Check the source to verify that this is the current version before use.

4.6.9 Insulation resistance. Resistors shall be tested in accordance with MIL-STD-202-302 (see 3.16). The following details and exceptions shall apply:

a. Special preparations: As specified in 4.6.8.1a.
b. Test condition letter: A or B, whichever is practicable.
c. Points of measurement: Between the resistor terminals connected together and the V-block.

4.6.10 Moisture resistance. Resistors shall be tested in accordance with MIL-STD-202-106 (see 3.17). The following details and exceptions shall apply:

a. Mounting: Soldered by their leads to rigid mounts or terminal lugs. The spacing of the mounts or terminals lugs shall be such that the length of each resistor is approximately .375 inch when measured from the edge of the supporting terminal to the resistor body. One half of the sample units shall be covered with a V-shaped metal strap whose width is .750 inch or equal to the body length, whichever is less (see figure 1). The strap shall be made of a corrosion resistant metal and shall be kept in contact with the resistor body by supporting the body with a nonconducting, noncorrosive support whose width is less than that of the body and which will not act as a moisture trap. The mounting straps may be individual for each resistor or continuous for all resistors.
b. Polarization and loading voltage:
   (1) Polarization: During steps 1 to 6, inclusive, a 100 volt dc potential shall be applied only to those resistors which have a polarizing strap. This potential shall be applied with the positive lead connected to the resistor terminals tied together, and the negative lead connected to the polarizing straps.
   (2) Loading voltage: During the first 2 hours of steps 1 and 4, a dc test potential equivalent to 100 percent rated wattage, but not exceeding 500 volts dc shall be applied to those resistors which are mounted by their leads without the polarizing straps specified in 4.6.10a.
c. Subcycle: Step 7a shall be performed during any five of the first eight cycles only. All polarizing straps may be removed to perform step 7a and shall then be replaced prior to returning the resistors to the humidity chamber.
d. Final measurements: Upon completion of step 6 of the final cycle, the resistors shall be held at the high humidity condition and at a temperature of +25°C ±2°C for a period of 1 hour 30 minutes to 3 hours 30 minutes. The same straps used for polarizing the resistors may also be used for the dielectric withstanding voltage and insulation resistance tests. Resistors shall be removed from the chamber (no forced air drying allowed after removal from the moisture chamber) and within 2 hours, the dc resistance, dielectric withstanding voltage, and insulation resistance shall be measured in that order as specified in 4.6.2, 4.6.8, and 4.6.9 respectively. The sample units shall not be subjected to forced circulating air during the tests.
e. Examination after test: Resistors shall be examined for evidence of mechanical damage.

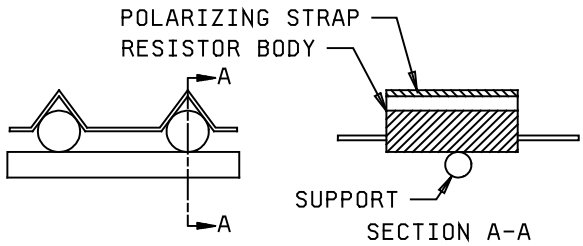


FIGURE 1. Mounting strap for moisture resistance test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:08Z
Check the source to verify that this is the current version before use.

4.6.11 Life. Resistors shall be tested in accordance with MIL-STD-202-108 (see 3.18). The following details and exceptions shall apply:

a. Method of mounting: Resistors shall be mounted and soldered on lightweight terminals. The effective length of each terminal shall be 1 inch. Resistors shall be arranged that the temperature of any one resistor shall not influence the temperature of any other resistor. There shall be no circulation of air over the resistors other than that caused by heat of the resistors.
b. Test temperature: +25°C ±5°C.
c. Initial measurement: DC resistance shall be measured at room temperature which shall be used as the reference reading for all subsequent measurements under the same conditions
d. Operating conditions: Rated dc continuous working voltage shall be applied intermittently, 45 minutes "on" and 15 minutes "off", for 1,000 hours.
e. Test condition: D.
f. Measurement during test: DC resistance shall be measured at the end of the 15 minute "off" periods, after, 250 hours +72 hours, -24 hours; 500 hours +72 hours, -24 hours; 1,000 hours +72 hours, -24 hours have elapsed. Measurement shall be made as near as possible to the specified time but may be adjusted so that measurement need not be made during other than normal workdays.
g. Examination after test: Resistors shall be examined for evidence of mechanical damage.

4.6.12 Resistance temperature characteristics. Resistors shall be tested in accordance with MIL-STD-202-304 (see 3.19). The following details and exceptions shall apply:

a. Reference temperature: Room ambient temperature.
b. Test temperature: In accordance with table X herein.
c. Accuracy of temperature measurement: Resistors shall be maintained for 30 minutes to 45 minutes within 1°C at each of the test temperatures as specified in table X herein. This tolerance shall be maintained at the established test temperatures.

TABLE X. Resistance temperature characteristic.

Sequence	Temperature °C 1/
	Group B inspection
1	Room temperature 2/
2	-55°C ±3°C
3	Room temperature 2/
4	+175°C ±3°C

1/ At the option of the manufacturer the reverse sequence may be as follows:

1	Room temperature
2	+175°C ±3°C
3	Room temperature
4	-55°C ±3°C

2/ This temperature shall be considered the reference temperature for each of the succeeding temperatures.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:08Z
Check the source to verify that this is the current version before use.

4.6.13 Shock (specified pulse). Resistors shall be tested in accordance with MIL-STD-202-213 (see 3.20). The following details and exceptions shall apply:

a. Special mounting means: Resistors shall be rigidly mounted on an appropriate jig fixtures with their leads supported at a distance of 0.25 inch from the resistor body. The resistor shall be mounted with body clamped or cemented to a flat surface. When used, the cement material shall not extend above the center line in the horizontal plane or beyond the resistor body in the vertical plane. In no case shall the resistor body be completely encapsulated. These fixtures shall be constructed in a manner to insure that the points of the resistor mounting supports will have the same motion as the shock table. Test leads used during this test shall be no larger than AWG size 22 stranded wire (not applicable for resistance value greater than 10 megohms). The test lead strength shall be no greater than is necessary. In all cases, the resistors shall be mounted in such a manner that the stress applied is in the direction which would be considered most detrimental.
b. Measurement before shock: DC resistance shall be measured as specified in 4.6.2.
c. Number and direction of applied shocks: The resistors shall be subjected to a total of 10 shocks in each of two mutually perpendicular planes, 1 perpendicular and the other parallel to the longitudinal axis of the resistor.
d. Test condition; I (100 G's, 6 ms sawtooth).
e. Measurement during shock (not applicable for resistance values greater than 10 megohms): Each resistor shall be monitored to determine electrical discontinuity by a method which shall at least be sensitive enough to monitor or register, automatically, any electrical discontinuity of 0.1 ms or greater duration.
f. Measurement after shock: DC resistance shall be examined for evidence of 4.6.2.
g. Examination after test: Resistors shall be examined for evidence of mechanical and electrical damage.

4.6.14 Vibration, high frequency. Resistors shall be tested in accordance with MIL-STD-202-204, (see 3.21). The following details and exceptions shall apply:

a. Mounting of specimens: Resistors shall be mounted on appropriate jig fixtures as specified in 4.6.13a. These fixtures shall be constructed in a manner to insure that the points of the resistor mounting supports will have the same motion as the vibration test table. The fixtures shall also be of a construction that will preclude any resonance in the fixture when subjected to vibration within the test range and the fixture shall be monitored for these features on the vibration table. A shielded cable containing test leads no larger than AWG 22 stranded wire shall be clamped to the resistor mounting fixture. The test lead length shall be no greater than necessary.
b. Initial measurement: DC resistance shall be measured as specified in 4.6.2.
c. Test condition: D (10 to 2,000 Hz, 20 G).
d. Direction of motion: In each of two mutually perpendicular planes, one perpendicular and the other parallel to the longitudinal axis of the resistor. Duration shall be 4 hours in each plane with total test time of 8 hours.
e. Measurements during vibration (not applicable to resistance values greater than 10 megohms): Each resistor shall be monitored to determine electrical discontinuity by a method which shall at least be sensitive enough to monitor or register, automatically, any electrical discontinuity of 0.1 ms or greater duration.
f. Measurement after vibration: DC resistance shall be measured as specified in 4.6.2.
g. Examination after test: Resistors shall be examined for evidence of mechanical damage.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:08Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The resistors covered by this specification are intended for use in electronic circuits where high voltages and high resistance are required.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification, the applicable associated specification, and complete PIN (see 1.2.1 and 3.1).
b. Unless otherwise specified (see 2.1), the versions of the individual documents referenced will be those in effect on the date of release of the solicitation.
c. Packaging requirements (see 5.1).
d. Whether the manufacturer performs the group C inspections or provides a certificate of compliance with group C requirements (see 4.5.1.2)

6.3 Qualification. Qualification is not required.

6.4 Tin whisker growth. The use of alloys with tin content greater than 97 percent, by mass, may exhibit tin whisker growth problems after manufacture. Tin whiskers may occur anytime from a day to years after manufacture and can develop under typical operating conditions, on products that use such materials. Conformal coatings applied over top of a whisker-prone surface will not prevent the formation of tin whiskers. Alloys of 3 percent lead, by mass, have shown to inhibit the growth of tin whiskers. For additional information on this matter, refer to ASTM-B545 (Standard Specification for Electrodeposited Coatings of Tin).

6.5 Application notes.

6.5.1 Voltage coefficient. Typical coefficients for high voltage resistors are 200 ppm/volt.

6.6 Subject term (key word) listing.

High resistance values
Axial leads
Maximum voltages

6.7 Pulse applications. Designers are CAUTIONED on using these resistors in high power pulse applications. Since they have not been qualified nor tested for such applications, damage and premature failure are possible.

6.8 Amendment notification. The margins of this specification are marked with vertical lines to indicate modification generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:08Z
Check the source to verify that this is the current version before use.

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5905-2026-001)

Review activities:
Army - AT, AV, CR4, MI
Navy - AS, CG, MC
Air Force - 19
Other - MDA

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:08Z
Check the source to verify that this is the current version before use.